4

Elementary plasma waves

4.1 General method for analyzing small-amplitude waves

All plasma phenomena can be described by combining Maxwell's equations with the Lorentz force equation where the latter is represented by the Vlasov, the two-fluid, or the MHD approximation. The subject of linear plasma waves provides a good introduction to the study of plasma phenomena because linear waves are relatively simple to analyze and yet demonstrate many essential features of plasma behavior.

Linear analysis, a straightforward method applicable to any set of partial differential equations describing a physical system, reveals the physical system's simplest non-trivial, self-consistent dynamical behavior. In the context of plasma dynamics, the method is as follows:

1. By making appropriate physical assumptions, the general Maxwell–Lorentz system of equations is reduced to the simplest set of equations characterizing the phenomena under consideration.
2. An equilibrium solution is determined for this set of equations. The equilibrium might be trivial such that densities are uniform, the plasma is neutral, and all velocities are zero. However, less trivial equilibria could also be invoked where there are density gradients or flow velocities. Equilibrium quantities are designated by the subscript 0, indicating "zero-order" in smallness.
3. If $\{f(\mathbf{x}, t), g(\mathbf{x}, t), h(\mathbf{x}, t), \ldots\}$ constitutes the set of dependent variables and a specific, physically allowed perturbation is prescribed for one of these variables, then solving the system of differential equations will give the responses of all the other dependent variables to this prescribed perturbation. For example, suppose that a perturbation ϵf_1 is prescribed for the dependent variable f where $\epsilon \ll 1$ so that f becomes

$$f = f_0 + \epsilon f_1 \tag{4.1}$$

and $\epsilon\,|f_1| \ll |f_0|$ for all $\mathbf{x}$ and t. The system of differential equations gives the functional dependence of the other variables on f and so, for example, would give $g = g(f) =$

$g(f_0+\epsilon f_1)$. Since the functional dependence of g on f is in general nonlinear, Taylor expansion gives $g = g_0+\epsilon g_1+\epsilon^2 g_2+\epsilon^3 g_3+\ldots$ The ϵ's are, from now on, considered implicit and so the variables are written as

$$\begin{aligned} f &= f_0+f_1 \\ g &= g_0+g_1+g_2+\ldots \\ h &= h_0+h_1+h_2+\ldots \end{aligned} \tag{4.2}$$

and it is assumed that the order of magnitude of f_1 is smaller than the order of magnitude of f_0 by a factor ϵ, etc. The smallness of the perturbation is an assumption that obviously must be satisfied in the real situation being modeled. Note that there are no f_j terms with $j \geq 2$ because the perturbation to f was prescribed as being f_1.

4. Each partial differential equation is rewritten with all dependent quantities expanded to first order as in Eq. (4.2). For example, the two-fluid continuity equation becomes

$$\frac{\partial(n_0+n_1)}{\partial t}+\nabla\cdot[(n_0+n_1)(\mathbf{u}_0+\mathbf{u}_1)] = 0. \tag{4.3}$$

By assumption, equilibrium quantities satisfy

$$\frac{\partial n_0}{\partial t}+\nabla\cdot(n_0\mathbf{u}_0) = 0. \tag{4.4}$$

The essence of linearization consists of subtracting the equilibrium equation (e.g., Eq. (4.4)) from the expanded equation (e.g., Eq. (4.3)). For this example such a subtraction yields

$$\frac{\partial n_1}{\partial t}+\nabla\cdot(n_1\mathbf{u}_0+n_0\mathbf{u}_1+n_1\mathbf{u}_1) = 0. \tag{4.5}$$

The nonlinear term $n_1\mathbf{u}_1$, which is a product of *two* first order quantities, is discarded because it is of order ϵ^2 whereas all the other terms are of order ϵ. What remains is called the linearized equation, i.e., the equation that consists of only first-order terms. For the example here, the linearized equation would be

$$\frac{\partial n_1}{\partial t}+\nabla\cdot(n_1\mathbf{u}_0+n_0\mathbf{u}_1) = 0. \tag{4.6}$$

The linearized equation is in a sense the differential of the original equation.

Before engaging in a methodical study of the large variety of waves that can propagate in a plasma, a few special cases of fundamental importance will first be examined.

4.2 Two-fluid theory of unmagnetized plasma waves

The simplest plasma waves are those described by two-fluid theory in an unmagnetized plasma, i.e., a plasma that has no equilibrium magnetic field. The theory for these waves also applies to magnetized plasmas in the special situation where all fluid motions are strictly parallel to the equilibrium magnetic field because fluid flowing along a magnetic field experiences no $\mathbf{u} \times \mathbf{B}$ force and so behaves as if there were no magnetic field.

The two-fluid equation of motion relevant to an unmagnetized plasma is

$$m_\sigma n_\sigma \frac{\mathrm{d}\mathbf{u}_\sigma}{\mathrm{d}t} = q_\sigma n_\sigma \mathbf{E} - \nabla P_\sigma \tag{4.7}$$

and these simple plasma waves are found by linearizing about an equilibrium where $\mathbf{u}_{\sigma 0} = 0$, $\mathbf{E}_0 = 0$, and $\nabla P_{\sigma 0} = 0$. The linearized form of Eq. (4.7) is then

$$m_\sigma n_{\sigma 0} \frac{\partial \mathbf{u}_{\sigma 1}}{\partial t} = q_\sigma n_{\sigma 0} \mathbf{E}_1 - \nabla P_{\sigma 1}. \tag{4.8}$$

The electric field can be expressed as

$$\mathbf{E} = -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t}, \tag{4.9}$$

a form that automatically satisfies Faraday's law. The vector potential $\mathbf{A}$ is undefined with respect to a gauge since $\mathbf{B} = \nabla \times (\mathbf{A} + \nabla\psi) = \nabla \times \mathbf{A}$. It is convenient to choose ψ so as to have $\nabla \cdot \mathbf{A} = 0$. This is called Coulomb gauge and causes the divergence of Eq. (4.9) to give Poisson's equation so that charge density provides the only source term for the electrostatic potential ϕ. Since Eq. (4.9) is linear to begin with, its linearized form is just

$$\mathbf{E}_1 = -\nabla\phi_1 - \frac{\partial \mathbf{A}_1}{\partial t}. \tag{4.10}$$

4.2.1 Electrostatic (compressional) waves

These waves are characterized by having finite $\nabla \cdot \mathbf{u}_1$ and are variously called compressional, electrostatic, or longitudinal waves. The first step in the analysis is to take the divergence of Eq. (4.8) to obtain

$$m_\sigma n_{\sigma 0} \frac{\partial \nabla \cdot \mathbf{u}_{\sigma 1}}{\partial t} = -q_\sigma n_{\sigma 0} \nabla^2 \phi_1 - \nabla^2 P_{\sigma 1}. \tag{4.11}$$

Because Eq. (4.11) involves three variables (i.e., $\mathbf{u}_{\sigma 1}$, ϕ_1, $P_{\sigma 1}$) two more equations are required to provide a complete description. One of these additional equations is the linearized continuity equation

$$\frac{\partial n_{\sigma 1}}{\partial t} + n_0 \nabla \cdot \mathbf{u}_{\sigma 1} = 0, \tag{4.12}$$

which, after substitution into Eq. (4.11), gives

$$m_\sigma \frac{\partial^2 n_{\sigma 1}}{\partial t^2} = q_\sigma n_{\sigma 0} \nabla^2 \phi_1 + \nabla^2 P_{\sigma 1}. \tag{4.13}$$

For adiabatic processes the pressure and density are related by

$$\frac{P_\sigma}{n_\sigma^\gamma} = const., \tag{4.14}$$

where $\gamma = (N+2)/N$ and N is the dimensionality of the system, whereas for isothermal processes

$$\frac{P_\sigma}{n_\sigma} = const. \tag{4.15}$$

The same formalism can therefore be used for both isothermal and adiabatic processes by using Eq. (4.14) for both and then simply setting $\gamma = 1$ if the process is isothermal. Linearization of Eq. (4.14) gives

$$\frac{P_{\sigma 1}}{P_{\sigma 0}} = \gamma \frac{n_{\sigma 1}}{n_{\sigma 0}} \tag{4.16}$$

so Eq. (4.13) becomes

$$m_\sigma \frac{\partial^2 n_{\sigma 1}}{\partial t^2} = q_\sigma n_{\sigma 0} \nabla^2 \phi_1 + \gamma \kappa T_{\sigma 0} \nabla^2 n_{\sigma 1}, \tag{4.17}$$

where $P_{\sigma 0} = n_{\sigma 0} \kappa T_{\sigma 0}$ has been used.

Although this system of linear equations could be solved by the formal method of Fourier transforms, we instead take the shortcut of simply assuming that the linear perturbation happens to be a single Fourier mode. Thus, it is assumed that *all* linearized dependent variables have the wave-like dependence

$$n_{\sigma 1} \sim \exp(\mathrm{i}\mathbf{k}\cdot\mathbf{x} - \mathrm{i}\omega t), \quad \phi_1 \sim \exp(\mathrm{i}\mathbf{k}\cdot\mathbf{x} - \mathrm{i}\omega t), \quad \text{etc.} \tag{4.18}$$

so that $\nabla \to \mathrm{i}\mathbf{k}$ and $\partial/\partial t \to -\mathrm{i}\omega$. Equation (4.17) therefore reduces to the algebraic equation

$$m_\sigma \omega^2 n_{\sigma 1} = q_\sigma n_{\sigma 0} k^2 \phi_1 + \gamma \kappa T_{\sigma 0} k^2 n_{\sigma 1}, \tag{4.19}$$

which may be solved for $n_{\sigma 1}$ to give

$$n_{\sigma 1} = \frac{q_\sigma n_{\sigma 0}}{m_\sigma} \frac{k^2 \phi_1}{(\omega^2 - \gamma k^2 \kappa T_{\sigma 0}/m_\sigma)}. \tag{4.20}$$

Poisson's equation provides another relation between ϕ_1 and $n_{\sigma 1}$, namely

$$k^2 \phi_1 = \frac{1}{\varepsilon_0} \sum_\sigma n_{\sigma 1} q_\sigma. \tag{4.21}$$

Equation (4.20) is substituted into Poisson's equation to give

$$k^2 \phi_1 = \sum_\sigma \frac{n_{\sigma 0} q_\sigma^2}{\varepsilon_0 m_\sigma} \frac{k^2 \phi_1}{(\omega^2 - \gamma k^2 \kappa T_{\sigma 0}/m_\sigma)}, \tag{4.22}$$

which may be rearranged as

$$\left[1-\sum_{\sigma}\frac{\omega_{p\sigma}^{2}}{(\omega^{2}-\gamma k^{2}\kappa T_{\sigma0}/m_{\sigma})}\right]\phi_{1}=0, \tag{4.23}$$

where

$$\omega_{p\sigma}^{2}\equiv\frac{n_{\sigma0}q_{\sigma}^{2}}{\varepsilon_{0}m_{\sigma}} \tag{4.24}$$

is the square of the *plasma frequency* of species σ. A useful way to recast Eq. (4.23) is

$$(1+\chi_{e}+\chi_{i})\phi_{1}=0, \tag{4.25}$$

where

$$\chi_{\sigma}=-\frac{\omega_{p\sigma}^{2}}{(\omega^{2}-\gamma k^{2}\kappa T_{\sigma0}/m_{\sigma})} \tag{4.26}$$

is called the susceptibility of species σ. In Eq. (4.25) the "1" comes from the "vacuum" part of Poisson's equation (i.e., the left-hand side term $\nabla^{2}\phi$) while the susceptibilities give the respective contributions of each species to the right-hand side of Poisson's equation. This formalism follows that of dielectrics where the displacement vector is $\mathbf{D}=\varepsilon\mathbf{E}$ and the dielectric constant is $\varepsilon=1+\chi$, where χ is a susceptibility.

Equation (4.25) shows that if $\phi_{1}\neq0,$ the quantity $1+\chi_{e}+\chi_{i}$ must vanish. In other words, in order to have a non-trivial normal mode it is necessary to have

$$1+\chi_{e}+\chi_{i}=0. \tag{4.27}$$

This is called a dispersion relation and prescribes a functional relation between ω and k. The dispersion relation can be considered as the determinant-like equation for the eigenvalues $\omega(k)$ of the system of equations.

The normal modes can be identified by noting that Eq. (4.26) has two limiting behaviors depending on how the wave-phase velocity compares to $\sqrt{\kappa T_{\sigma0}/m_{\sigma}}$, a quantity that is of the order of the thermal velocity. These limiting behaviors are:

1. Adiabatic regime: $\omega/k\gg\sqrt{\kappa T_{\sigma0}/m_{\sigma}}$ and $\gamma=(N+2)/N$. Because plane waves are one-dimensional perturbations (i.e., the plasma is compressed in the $\hat{k}$ direction only), $N=1$ so that $\gamma=3$. Hence the susceptibility has the limiting form

$$\begin{aligned}\chi_{\sigma}&=-\frac{\omega_{p\sigma}^{2}}{\omega^{2}(1-\gamma k^{2}\kappa T_{\sigma0}/m_{\sigma}\omega^{2})}\\&\simeq-\frac{\omega_{p\sigma}^{2}}{\omega^{2}}\left(1+3\frac{k^{2}}{\omega^{2}}\frac{\kappa T_{\sigma0}}{m_{\sigma}}\right)\\&=-\frac{1}{k^{2}\lambda_{D\sigma}^{2}}\frac{k^{2}}{\omega^{2}}\frac{\kappa T_{\sigma}}{m_{\sigma}}\left(1+3\frac{k^{2}}{\omega^{2}}\frac{\kappa T_{\sigma0}}{m_{\sigma}}\right).\end{aligned} \tag{4.28}$$

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2. Isothermal regime: $\omega/k \ll \sqrt{\kappa T_{\sigma 0}/m_\sigma}$ and $\gamma = 1$. Here the susceptibility has the limiting form

$$\chi_\sigma = \frac{\omega_{p\sigma}^2}{k^2 \kappa T_{\sigma 0}/m_\sigma} = \frac{1}{k^2 \lambda_{D\sigma}^2}. \tag{4.29}$$

Figure 4.1 shows a plot of χ_σ versus $\omega/k\sqrt{\kappa T_{\sigma 0}/m_\sigma}$. The isothermal and adiabatic susceptibilities are seen to be substantially different and, in particular, do not coalesce when $\omega/k\sqrt{\kappa T_{\sigma 0}/m_\sigma} \to 1$. This non-coalescence as $\omega/k\sqrt{\kappa T_{\sigma 0}/m_\sigma} \to 1$ indicates that the fluid description, while valid in both the adiabatic and isothermal limits, fails in the vicinity of $\omega/k \sim \sqrt{\kappa T_{\sigma 0}/m_\sigma}$. As will be seen later, the more accurate Vlasov description must be used in the $\omega/k \sim \sqrt{\kappa T_{\sigma 0}/m_\sigma}$ regime.

Since the ion-to-electron mass ratio is large, ions and electrons typically have thermal velocities differing by at least one and sometimes two orders of magnitude. Furthermore, ion and electron temperatures often differ, again allowing substantially different electron and ion thermal velocities. Three different situations can occur in a typical plasma depending on how the wave-phase velocity compares to thermal velocities. These situations are:

1. Case where $\omega/k \gg \sqrt{\kappa T_{e0}/m_e}, \sqrt{\kappa T_{i0}/m_i}$.
   Here both electrons and ions are adiabatic and the dispersion relation becomes

$$1 - \frac{\omega_{pe}^2}{\omega^2}\left(1 + 3\frac{k^2}{\omega^2}\frac{\kappa T_{e0}}{m_e}\right) - \frac{\omega_{pi}^2}{\omega^2}\left(1 + 3\frac{k^2}{\omega^2}\frac{\kappa T_{i0}}{m_i}\right) = 0. \tag{4.30}$$

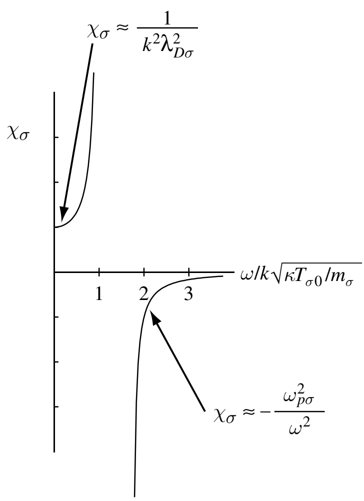

Fig. 4.1 Susceptibility χ_σ as a function of $\omega/k\sqrt{\kappa T_{\sigma 0}/m_\sigma}$.

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Since $\omega_{pe}^2/\omega_{pi}^2 = m_i/m_e$ the ion contribution can be dropped, and the dispersion becomes

$$1 - \frac{\omega_{pe}^2}{\omega^2}\left(1 + 3\frac{k^2}{\omega^2}\frac{\kappa T_{e0}}{m_e}\right) = 0. \tag{4.31}$$

To lowest order, the solution of this equation is simply $\omega^2 = \omega_{pe}^2$. An iterative solution may be obtained by substituting this lowest order solution into the thermal term, which, by assumption, is a small correction because $\omega/k \gg \sqrt{\kappa T_{e0}/m_e}$. This gives the standard form for the high-frequency, electrostatic, unmagnetized plasma wave

$$\omega^2 = \omega_{pe}^2 + 3k^2\frac{\kappa T_{e0}}{m_e}. \tag{4.32}$$

This most basic of plasma waves is called the electron plasma wave, the Langmuir wave (Langmuir 1928), or the Bohm–Gross wave (Bohm and Gross 1949).

2. Case where $\omega/k \ll \sqrt{\kappa T_{e0}/m_e}, \sqrt{\kappa T_{i0}/m_i}$.
Here both electrons and ions are isothermal and the dispersion becomes

$$1 + \sum_\sigma \frac{1}{k^2\lambda_{D\sigma}^2} = 0. \tag{4.33}$$

This has no frequency dependence, and is just the Debye shielding derived in Chapter 1. Thus, when $\omega/k \ll \sqrt{\kappa T_{e0}/m_e}, \sqrt{\kappa T_{i0}/m_i}$ the plasma approaches the steady-state limit and screens out any applied perturbation. This limit shows why ions cannot provide Debye shielding for electrons, because if the test particle were chosen to be an electron, its nominal speed would be the electron thermal velocity. From the point of view of ions, this fast-moving electron test particle would zoom through with a phase velocity $\omega/k \sim v_{Te}$ and so violate the requirement that $\omega/k \ll \sqrt{\kappa T_{i0}/m_i}$ for the ions to be able to respond; the ions would not be able to move fast enough to do any shielding (Wang, Joyce, and Nicholson 1981).

3. Case where $\sqrt{\kappa T_{i0}/m_i} \ll \omega/k \ll \sqrt{\kappa T_{e0}/m_e}$.
Here the ions act adiabatically whereas the electrons act isothermally so that the dispersion becomes

$$1 + \frac{1}{k^2\lambda_{De}^2} - \frac{\omega_{pi}^2}{\omega^2}\left(1 + 3\frac{k^2}{\omega^2}\frac{\kappa T_{i0}}{m_i}\right) = 0. \tag{4.34}$$

It is conventional to define the "ion acoustic" velocity

$$c_s^2 = \omega_{pi}^2\lambda_{De}^2 = \kappa T_e/m_i \tag{4.35}$$

so that Eq. (4.34) can be recast as

$$\omega^2 = \frac{k^2 c_s^2}{1 + k^2\lambda_{De}^2}\left(1 + 3\frac{k^2}{\omega^2}\frac{\kappa T_{i0}}{m_i}\right). \tag{4.36}$$

Since $\omega/k \gg \sqrt{\kappa T_{i0}/m_i}$, this may be solved iteratively by first assuming $T_{i0} = 0$ giving

$$\omega^2 = \frac{k^2 c_s^2}{1 + k^2\lambda_{De}^2}. \tag{4.37}$$

This is the most basic form for the *ion acoustic wave* dispersion and in the limit $k^2\lambda_{De}^2 \gg 1$, becomes simply $\omega^2 = c_s^2/\lambda_{De}^2 = \omega_{pi}^2$. To obtain the next higher order of precision for the ion acoustic dispersion, Eq. (4.37) may be used to eliminate k^2/ω^2 from the ion thermal term of Eq. (4.36) giving

$$\omega^2 = \frac{k^2 c_s^2}{1+k^2\lambda_{De}^2} + 3k^2\frac{\kappa T_{i0}}{m_i}. \tag{4.38}$$

For self-consistency, it is necessary to have $c_s^2 \gg \kappa T_{i0}/m_i$; if this were not true, the ion acoustic wave would become $\omega^2 = 3k^2\kappa T_{i0}/m_i$, which would violate the assumption that $\omega/k \gg \sqrt{\kappa T_{i0}/m_i}$. The condition $c_s^2 \gg \kappa T_{i0}/m_i$ is the same as $T_e \gg T_i$ so ion acoustic waves can only propagate when the electrons are much hotter than the ions. This issue will be explored in more depth later when ion acoustic waves are re-examined from the Vlasov point of view.

4.2.2 Electromagnetic (incompressible) waves

The compressional waves discussed in the previous section were obtained by taking the divergence of Eq. (4.8). An arbitrary vector field $\mathbf{V}$ can always be decomposed into a gradient of a potential and a solenoidal part, i.e., it can always be written as $\mathbf{V} = \nabla\psi + \nabla\times\mathbf{Q}$, where ψ and $\mathbf{Q}$ can be determined from $\mathbf{V}$. The $\nabla\psi$ term has zero curl and so describes a conservative field, whereas the solenoidal term $\nabla\times\mathbf{Q}$ has zero divergence and describes a non-conservative field. Because we have chosen to use Coulomb gauge in our analysis of waves, the $-\nabla\phi$ term on the right-hand side of Eq. (4.9) is the only conservative field; the $-\partial\mathbf{A}/\partial t$ term is the solenoidal or non-conservative field.

Waves involving finite $\mathbf{A}$ have coupled electric and magnetic fields and are a generalization of vacuum electromagnetic waves such as light or radio waves. These finite $\mathbf{A}$ waves are variously called electromagnetic, transverse, or incompressible waves. Since no electrostatic potential is involved, $\nabla\cdot\mathbf{E} = 0$ and the plasma remains neutral. Because $\mathbf{A} \neq 0$, these waves involve electric currents.

Since the electromagnetic waves are solenoidal, the $-\nabla\phi$ term in Eq. (4.8) is superfluous and can be eliminated by taking the curl of Eq. (4.8) giving

$$\frac{\partial}{\partial t}\nabla\times(m_\sigma n_\sigma \mathbf{u}_{\sigma 1}) = -q_\sigma n_\sigma \frac{\partial \mathbf{B}_1}{\partial t}. \tag{4.39}$$

To obtain an equation involving currents, Eq. (4.39) is integrated with respect to time, multiplied by q_σ/m_σ, and then summed over species to give

$$\nabla\times\mathbf{J}_1 = -\varepsilon_0\omega_p^2\mathbf{B}_1, \tag{4.40}$$

where

$$\omega_p^2 = \sum_\sigma \omega_{p\sigma}^2. \tag{4.41}$$

However, Ampère's law can be written in the form

$$\mathbf{J}_1 = \frac{1}{\mu_0}\nabla\times\mathbf{B}_1 - \varepsilon_0\frac{\partial\mathbf{E}_1}{\partial t}, \tag{4.42}$$

which, after substitution into Eq. (4.40), gives

$$\nabla\times\left(\nabla\times\mathbf{B}_1 - \frac{1}{c^2}\frac{\partial\mathbf{E}_1}{\partial t}\right) = -\frac{\omega_p^2}{c^2}\mathbf{B}_1. \tag{4.43}$$

Using the vector identity $\nabla\times(\nabla\times\mathbf{Q}) = \nabla(\nabla\cdot\mathbf{Q}) - \nabla^2\mathbf{Q}$ and Faraday's law this becomes

$$\nabla^2\mathbf{B}_1 = \frac{1}{c^2}\frac{\partial^2\mathbf{B}_1}{\partial t^2} + \frac{\omega_p^2}{c^2}\mathbf{B}_1. \tag{4.44}$$

In the limit of no plasma, $\omega_p^2 \to 0$, so that Eq. (4.44) reduces to the standard vacuum electromagnetic wave. If it is assumed that $\mathbf{B}_1 \sim \exp(\mathrm{i}\mathbf{k}\cdot\mathbf{x} - \mathrm{i}\omega t)$, Eq. (4.44) becomes the electromagnetic, unmagnetized plasma wave dispersion

$$\omega^2 = \omega_p^2 + k^2c^2. \tag{4.45}$$

Waves satisfying Eq. (4.45) are often used to measure plasma density. Such a measurement can be accomplished two ways:

1. Cutoff method
   If $\omega^2 < \omega_p^2$ then k^2 becomes negative, the wave does not propagate, and only exponentially growing or decaying spatial behavior occurs (such behavior is called evanescent). If the wave is excited by an antenna driven by a fixed-frequency oscillator, the boundary condition that the wave field does not diverge at infinity means that only waves that decay away from the antenna exist. Thus, the field is localized near the antenna and there is no wave-like behavior. This is called *cutoff*. When the oscillator frequency is raised above ω_p, the wave starts to propagate so that a receiver located some distance away will abruptly start to pick up the wave. By scanning the transmitter frequency and noting the frequency at which the wave starts to propagate, ω_p^2 is determined, giving a direct, unambiguous measurement of the plasma density.

2. Phase shift method
   Here the oscillator frequency is set to be well above cutoff so that the wave is always propagating. The dispersion relation is solved for k and the phase delay $\Delta\phi$ of the wave through the plasma is measured by interferometric fringe-counting. The total phase delay through a length L of plasma is

$$\phi = \int_0^L k\mathrm{d}x = \frac{1}{c}\int_0^L \left[\omega^2 - \omega_p^2\right]^{1/2}\mathrm{d}x \simeq \frac{\omega}{c}\int_0^L \left[1 - \frac{\omega_p^2}{2\omega^2}\right]\mathrm{d}x \tag{4.46}$$

   so that the phase delay due to the presence of plasma is

$$\Delta\phi = -\frac{1}{2\omega c}\int_0^L \omega_{p_e}^2\mathrm{d}x = -\frac{e^2}{2\omega c m_e\varepsilon_0}\int_0^L n\mathrm{d}x. \tag{4.47}$$

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Thus, measurement of the phase shift $\Delta\phi$ due to the presence of plasma can be used to measure the average density along L; this density is called the *line-averaged density*.

4.3 Low-frequency magnetized plasma: Alfvén waves

4.3.1 Overview of Alfvén waves

We now consider low-frequency waves propagating in a *magnetized* plasma, i.e., a plasma immersed in a uniform, constant magnetic field $\mathbf{B} = B_0\hat{z}$. By low frequency, it is meant that the wave frequency ω is much smaller than the ion cyclotron frequency ω_{ci}. Several types of waves exist in this frequency range; certain of these involve electric fields having a purely electrostatic character (i.e., $\nabla\times\mathbf{E} = 0$), whereas others involve electric fields having an inductive character (i.e., $\nabla\times\mathbf{E} \neq 0$). Faraday's law, $\nabla\times\mathbf{E} = -\partial\mathbf{B}/\partial t$, shows that if the electric field is electrostatic the magnetic field must be constant, whereas inductive electric fields must have an associated time-dependent magnetic field.

We now further restrict attention to a specific category of these $\omega \ll \omega_{ci}$ modes, namely the Alfvén waves. These waves are the normal modes of MHD, involve magnetic perturbations, and have characteristic velocities of the order of the Alfvén velocity $v_A = B/\sqrt{\mu_0\rho}$. The existence of such modes is not surprising if one considers that ordinary neutral gas sound waves have a velocity $c_{sound} = \sqrt{\gamma P/\rho}$ and the magnetic stress tensor scales as $\sim B^2/\mu_0\rho$ so that Alfvén-type velocities will result if P is replaced by $B^2/2\mu_0$. Two distinct kinds of Alfvén modes exist and to complicate matters these are called a variety of names by different authors. One mode, variously called the fast mode, the compressional mode, or the magnetosonic mode resembles a sound wave and involves compression and rarefaction of magnetic field lines; this mode has a finite B_{z1}. The other mode, variously called the Alfvén mode, the shear mode, the torsional mode, or the slow mode, involves twisting, shearing, or plucking motions; this mode has $B_{z1} = 0$. This latter mode appears in distinct β-dependent versions when modeled using two-fluid or Vlasov theory; these are respectively called the inertial Alfvén wave and the kinetic Alfvén wave.

Both Faraday's law and the pre-Maxwell Ampère's law are fundamental to Alfvén wave dynamics. The system of linearized equations thus is

$$\nabla\times\mathbf{E}_1 = -\frac{\partial\mathbf{B}_1}{\partial t} \tag{4.48}$$

$$\nabla\times\mathbf{B}_1 = \mu_0\mathbf{J}_1. \tag{4.49}$$

If the dependence of $\mathbf{J}_1$ on $\mathbf{E}_1$ can be determined, then the combination of Ampère's law and Faraday's law provides a complete self-consistent description of the coupled fields $\mathbf{E}_1, \mathbf{B}_1$ and hence describes the normal modes. From a

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mathematical point of view, specifying $\mathbf{J}_1(\mathbf{E}_1)$ means that there are as many equations as dependent variables in Eqs. (4.48) and (4.49). The relationship between $\mathbf{J}_1$ and $\mathbf{E}_1$ is determined by the Lorentz equation or some generalization thereof (e.g., drift equations, Vlasov equation, fluid equation of motion). The analysis of solutions to Eqs. (4.48) and (4.49) will proceed in three steps. First, we will consider the zero-pressure MHD approximation, next, the finite-pressure MHD approximation, and finally, the finite-pressure two-fluid approximation. MHD is a simpler description than two-fluid theory because MHD essentially ignores parallel dynamics. Instead of attempting a description of the parallel dynamics, the MHD approximation invokes what is in effect an ad hoc closure relation for the parallel current density in order to maintain overall charge neutrality.

4.3.2 Zero-pressure MHD model

The zero-pressure MHD model reveals the fundamental nature of the Alfvén modes and shows that the essence of these MHD modes comes from the polarization drift associated with a time-dependent perpendicular electric field, namely

$$\mathbf{u}_{\sigma,polarization} = \frac{m_\sigma}{q_\sigma B^2}\frac{\mathrm{d}\mathbf{E}_\perp}{\mathrm{d}t}; \tag{4.50}$$

this was discussed in the derivation of Eq. (3.77). The polarization drift results in a polarization current

$$\begin{aligned}\mathbf{J}_\perp &= \sum n_\sigma q_\sigma \mathbf{u}_{\sigma,polarization}\\ &= \frac{\rho}{B^2}\frac{\mathrm{d}\mathbf{E}_\perp}{\mathrm{d}t},\end{aligned} \tag{4.51}$$

where $\rho = \sum n_\sigma m_\sigma$ is the mass density. This can be recast as

$$\begin{aligned}\frac{\mathrm{d}\mathbf{E}_\perp}{\mathrm{d}t} &= \frac{B^2}{\mu_0\rho}\mu_0\mathbf{J}_\perp\\ &= v_A^2\left(\nabla\times\mathbf{B}_1\right)_\perp,\end{aligned} \tag{4.52}$$

where

$$v_A^2 = \frac{B^2}{\mu_0\rho} \tag{4.53}$$

is the Alfvén velocity. Equation (4.48) and the linearized version of Eq. (4.52) give the two basic coupled equations governing these Alfvén modes, namely

$$\frac{\partial\mathbf{B}_1}{\partial t} = -\nabla\times\mathbf{E}_1 \tag{4.54a}$$

$$\frac{\partial\mathbf{E}_{\perp 1}}{\partial t} = v_A^2\left(\nabla\times\mathbf{B}_1\right)_\perp. \tag{4.54b}$$

The omission of an equation for the parallel component of Ampère's law is an essential property of the MHD model. Unlike all other dependent variables, which appear in two equations and so have to be solved for self-consistently, the only place where J_z appears in MHD is in the $\nabla\cdot\mathbf{J}=0$ equation and so J_z is solved for in MHD as a sort of "after-thought" using the relation $J_z=-\int \mathrm{d}z\nabla_\perp\cdot\mathbf{J}_\perp$.

A critical feature of MHD related to this omission of a dynamical prescription for J_z results from dotting the linearized ideal Ohm's law

$$\mathbf{E}_1+\mathbf{U}_1\times\mathbf{B}=0 \tag{4.55}$$

with $\hat{z}$ to obtain $E_{z1}=0$. Thus, in the MHD limit, it is assumed there is no parallel electric field so there is no means for accelerating particles in the parallel direction and thus no means for attaining the parallel particle velocities associated with the existence of a parallel current. Even though MHD does not provide the dynamics for establishing a parallel current, MHD nevertheless requires the existence of a parallel current in order to satisfy $\nabla\cdot\mathbf{J}=0$. Thus, according to the MHD approximation, a parallel current spontaneously comes into existence without any parallel electric field to drive this current. Equation (4.55) also implies that the plasma velocity for these modes is just the $\mathbf{E}\times\mathbf{B}$ drift, i.e., $\mathbf{U}_1=\mathbf{E}_1\times\mathbf{B}/B^2$.

We now proceed with the derivation of the two zero-pressure MHD modes. Since E_{z1} is zero, the electric field is only in the perpendicular direction. The components of the electric field, the magnetic field, and the ∇ operator can thus be expressed as

$$\begin{aligned}\mathbf{E}_1&=\mathbf{E}_{\perp 1}\\ \mathbf{B}_1&=\mathbf{B}_{\perp 1}+B_{z1}\hat{z}\\ \nabla&=\nabla_\perp+\hat{z}\frac{\partial}{\partial z},\end{aligned} \tag{4.56}$$

where $\perp$ refers to the x and y components of a vector or a vector operator so, for example, $\nabla_\perp=\hat{x}\partial/\partial x+\hat{y}\partial/\partial y$. The curl operators appearing in Eqs. (4.54) can therefore be expanded as

$$\begin{aligned}\nabla\times\mathbf{E}_1&=\left(\nabla_\perp+\hat{z}\frac{\partial}{\partial z}\right)\times\mathbf{E}_{\perp 1}\\ &=\nabla_\perp\times\mathbf{E}_{\perp 1}+\hat{z}\times\frac{\partial\mathbf{E}_{\perp 1}}{\partial z}\end{aligned} \tag{4.57}$$

and

$$\begin{aligned}(\nabla\times\mathbf{B}_1)_\perp&=\left(\left(\nabla_\perp+\hat{z}\frac{\partial}{\partial z}\right)\times\left(\mathbf{B}_{\perp 1}+B_{z1}\hat{z}\right)\right)_\perp\\ &=\nabla_\perp B_{z1}\times\hat{z}+\hat{z}\times\frac{\partial\mathbf{B}_{\perp 1}}{\partial z},\end{aligned} \tag{4.58}$$

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where it should be noted that both $\nabla_\perp \times \mathbf{E}_{\perp 1}$ and $\nabla_\perp \times \mathbf{B}_{\perp 1}$ are in the z direction. Using Eqs. (4.57) and (4.58) in Eqs. (4.54) gives the basic pair of coupled equations for MHD waves, namely

$$\frac{\partial \mathbf{B}_1}{\partial t} = -\nabla_\perp \times \mathbf{E}_{\perp 1} - \hat{z} \times \frac{\partial \mathbf{E}_{\perp 1}}{\partial z} \tag{4.59a}$$

$$\frac{\partial \mathbf{E}_{\perp 1}}{\partial t} = v_A^2 \left(\nabla_\perp B_{z1} \times \hat{z} + \hat{z} \times \frac{\partial \mathbf{B}_{\perp 1}}{\partial z} \right). \tag{4.59b}$$

Slow or Alfvén mode (mode where $B_{z1} = 0$)

In this case $\mathbf{B}_1 = \mathbf{B}_{\perp 1}$ and Eqs. (4.59) become

$$\frac{\partial \mathbf{B}_{\perp 1}}{\partial t} = -\hat{z} \times \frac{\partial \mathbf{E}_{\perp 1}}{\partial z} \tag{4.60a}$$

$$\frac{\partial \mathbf{E}_{\perp 1}}{\partial t} = v_A^2 \hat{z} \times \frac{\partial \mathbf{B}_{\perp 1}}{\partial z} \tag{4.60b}$$

with the condition

$$\nabla_\perp \times \mathbf{E}_{\perp 1} = 0, \tag{4.61}$$

which is obtained from the z component of Eq. (4.59a). Equations (4.60) can be rewritten as

$$\frac{\partial \mathbf{B}_{\perp 1}}{\partial t} = \frac{\partial}{\partial z} \left(\hat{z} \times \mathbf{E}_{\perp 1} \right) \tag{4.62a}$$

$$\frac{\partial}{\partial t} \left(\hat{z} \times \mathbf{E}_{\perp 1} \right) = -v_A^2 \frac{\partial \mathbf{B}_{\perp 1}}{\partial z}, \tag{4.62b}$$

which leads to a wave equation in the coupled variables $\hat{z} \times \mathbf{E}_{\perp 1}$ and $\mathbf{B}_{\perp 1}$. In particular, taking a second time derivative of Eq. (4.62a) and then substituting for $\partial/\partial t \left(\hat{z} \times \mathbf{E}_{\perp 1} \right)$ using Eq. (4.62b), gives the wave equation for the slow mode (Alfvén mode),

$$\frac{\partial^2 \mathbf{B}_{\perp 1}}{\partial t^2} = v_A^2 \frac{\partial^2 \mathbf{B}_{\perp 1}}{\partial z^2}. \tag{4.63}$$

This is the mode originally derived by Alfvén (1943). The condition given by Eq. (4.61) corresponds to setting $B_{z1} = 0$ and implies that $\mathbf{E}_{\perp 1}$ can be expressed as the gradient of a potential. This further implies that the velocity $\mathbf{U}_1 = \mathbf{E}_1 \times \mathbf{B}/B^2$ is incompressible since $\nabla \cdot \mathbf{U}_1 = \nabla \cdot \left(\mathbf{E}_1 \times \mathbf{B}/B^2 \right) = B^{-2}\mathbf{B} \cdot \nabla \times \mathbf{E}_1 = B^{-2}\mathbf{B} \cdot \nabla_\perp \times \mathbf{E}_{\perp 1} = 0$.

4.3.3 Fast mode (mode where $B_{z1} \neq 0$)

In this case only the z component of Eq. (4.59a) is used and after crossing Eq. (4.59b) with $\hat{z}$, Eqs. (4.59) become

$$\frac{\partial B_{z1}}{\partial t} = -\hat{z} \cdot \nabla_\perp \times \mathbf{E}_{\perp 1} = -\nabla \cdot (\mathbf{E}_{\perp 1} \times \hat{z}) \tag{4.64a}$$

$$\frac{\partial}{\partial t}\mathbf{E}_{\perp 1} \times \hat{z} = v_A^2 \left(\nabla_\perp B_{z1} \times \hat{z} + \hat{z} \times \frac{\partial \mathbf{B}_{\perp 1}}{\partial z} \right) \times \hat{z} = v_A^2 \left(-\nabla_\perp B_{z1} + \frac{\partial \mathbf{B}_{\perp 1}}{\partial z} \right). \tag{4.64b}$$

Taking a time derivative of Eq. (4.64a) and then using Eq. (4.64b) to substitute for $\partial/\partial t\,(\mathbf{E}_{\perp 1} \times \hat{z})$ gives

$$\begin{aligned} \frac{\partial^2 B_{z1}}{\partial t^2} &= -v_A^2 \nabla \cdot \left(-\nabla_\perp B_{z1} + \frac{\partial \mathbf{B}_{\perp 1}}{\partial z} \right) \\ &= v_A^2 \nabla_\perp^2 B_{z1} - v_A^2 \frac{\partial}{\partial z} (\nabla \cdot \mathbf{B}_{\perp 1}). \end{aligned} \tag{4.65}$$

However, using $\nabla \cdot \mathbf{B}_1 = 0$ it is seen that

$$\nabla \cdot \mathbf{B}_{\perp 1} = -\frac{\partial B_{z1}}{\partial z} \tag{4.66}$$

and so the fast wave equation becomes

$$\frac{\partial^2 B_{z1}}{\partial t^2} = v_A^2 \nabla^2 B_{z1}. \tag{4.67}$$

4.3.4 Comparison of the two modes

The slow mode Eq. (4.63) involves only z derivatives and so has a dispersion relation

$$\omega^2 = k_z^2 v_A^2, \tag{4.68}$$

whereas the fast mode involves the ∇^2 operator and so has the dispersion relation

$$\omega^2 = k^2 v_A^2. \tag{4.69}$$

The slow mode is incompressible and has $B_{z1} = 0$ so its perturbed magnetic field is entirely orthogonal to the equilibrium field. Thus, the slow mode magnetic perturbation corresponds to a twisting or plucking of the equilibrium field. The fast mode has $B_{z1} \neq 0$, which corresponds to a compression of the equilibrium field as shown in Fig. 4.2; since the plasma is frozen to the magnetic field, the plasma is also compressed.

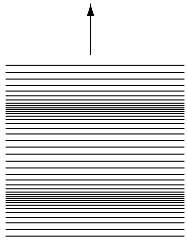

Fig. 4.2 Compressional Alfvén waves.

4.3.5 Finite-pressure analysis of MHD waves

If the pressure is allowed to be finite, then the two modes become coupled and an acoustic mode appears. Using the vector identity $\nabla B^2 = 2(\mathbf{B}\cdot\nabla\mathbf{B} + \mathbf{B}\times\nabla\times\mathbf{B})$ the $\mathbf{J}\times\mathbf{B}$ force in the MHD equation of motion can be written as

$$\mathbf{J}\times\mathbf{B} = \frac{1}{\mu_0}(\nabla\times\mathbf{B})\times\mathbf{B} - \nabla\left(\frac{B^2}{2\mu_0}\right) + \frac{1}{\mu_0}\mathbf{B}\cdot\nabla\mathbf{B}. \tag{4.70}$$

The MHD equation of motion thus becomes

$$\rho\frac{\mathrm{D}\mathbf{U}}{\mathrm{D}t} = -\nabla\left(P + \frac{B^2}{2\mu_0}\right) + \frac{1}{\mu_0}\mathbf{B}\cdot\nabla\mathbf{B}. \tag{4.71}$$

Linearizing this equation about a stationary equilibrium, where the pressure and the density are uniform and constant, gives

$$\rho\frac{\partial\mathbf{U}_1}{\partial t} = -\nabla\left(P_1 + \frac{\mathbf{B}\cdot\mathbf{B}_1}{\mu_0}\right) + \frac{1}{\mu_0}\mathbf{B}\cdot\nabla\mathbf{B}_1, \tag{4.72}$$

where unsubscripted dependent variables are equilibrium (zero-order) quantities. The curl of the linearized ideal MHD Ohm's law,

$$\mathbf{E}_1 + \mathbf{U}_1\times\mathbf{B} = 0, \tag{4.73}$$

gives the induction equation

$$\frac{\partial\mathbf{B}_1}{\partial t} = \nabla\times(\mathbf{U}_1\times\mathbf{B}), \tag{4.74}$$

while the linearized continuity equation

$$\frac{\partial\rho_1}{\partial t} + \rho\nabla\cdot\mathbf{U}_1 = 0 \tag{4.75}$$

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together with the equation of state

$$\frac{P_1}{P} = \gamma \frac{\rho_1}{\rho} \tag{4.76}$$

give

$$\frac{\partial P_1}{\partial t} = -\gamma P \nabla \cdot \mathbf{U}_1. \tag{4.77}$$

To obtain an equation involving $\mathbf{U}_1$ only, we take the time derivative of Eq. (4.72) and use Eqs. (4.74) and (4.77) to eliminate the time derivatives of P_1 and $\mathbf{B}_1$. This gives

$$\begin{aligned} \rho \frac{\partial^2 \mathbf{U}_1}{\partial t^2} = & -\nabla \left(-\gamma P \nabla \cdot \mathbf{U}_1 + \frac{1}{\mu_0} \mathbf{B} \cdot \nabla \times (\mathbf{U}_1 \times \mathbf{B}) \right) \\ & + \frac{1}{\mu_0} (\mathbf{B} \cdot \nabla) \nabla \times (\mathbf{U}_1 \times \mathbf{B}) . \end{aligned} \tag{4.78}$$

This can be simplified using the identity $\nabla \cdot (\mathbf{a} \times \mathbf{b}) = \mathbf{b} \cdot \nabla \times \mathbf{a} - \mathbf{a} \cdot \nabla \times \mathbf{b}$ so that

$$\mathbf{B} \cdot \nabla \times (\mathbf{U}_1 \times \mathbf{B}) = \nabla \cdot [(\mathbf{U}_1 \times \mathbf{B}) \times \mathbf{B}] = -B^2 \nabla \cdot \mathbf{U}_{1\perp}. \tag{4.79}$$

Furthermore,

$$\mathbf{B} \cdot \nabla = B \frac{\partial}{\partial z} = \mathrm{i} k_z B. \tag{4.80}$$

Using these relations Eq. (4.78) becomes

$$\frac{\partial^2 \mathbf{U}_1}{\partial t^2} = \nabla \left(c_s^2 \nabla \cdot \mathbf{U}_1 + v_A^2 \nabla \cdot \mathbf{U}_{1\perp} \right) + \mathrm{i} k_z v_A^2 \nabla \times (\mathbf{U}_1 \times \hat{z}), \tag{4.81}$$

where $c_s^2 = \gamma P / \rho$ is the velocity of the *conventional sound wave* in a gas (this wave is not to be confused with the ion acoustic wave, which is a result of two-fluid analysis and which has a velocity $c_s^2 = \kappa T_e / m_i$). To proceed further we take either the divergence or the curl of this equation to obtain expressions for compressional or incompressible motions.

4.3.6 MHD compressional (fast) mode

Here we take the divergence of Eq. (4.81) to obtain

$$\frac{\partial^2 \nabla \cdot \mathbf{U}_1}{\partial t^2} = \nabla^2 \left(c_s^2 \nabla \cdot \mathbf{U}_1 + v_A^2 \nabla \cdot \mathbf{U}_{1\perp} \right) \tag{4.82}$$

or

$$\omega^2 \nabla \cdot \mathbf{U}_1 = \left(k_\perp^2 + k_z^2 \right) \left(c_s^2 \nabla \cdot \mathbf{U}_1 + v_A^2 \nabla \cdot \mathbf{U}_{1\perp} \right). \tag{4.83}$$

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On the other hand, if Eq. (4.81) is operated on with $\nabla_\perp = \nabla - \mathrm{i}k_z\hat{z}$ we obtain

$$\frac{\partial^2 \nabla_\perp \cdot U_1}{\partial t^2} = \nabla_\perp^2 \left(c_s^2 \nabla \cdot \mathbf{U}_1 + v_A^2 \nabla \cdot \mathbf{U}_{1\perp}\right) + k_z^2 v_A^2 \hat{z} \cdot \nabla \times (\mathbf{U}_1 \times \hat{z}). \tag{4.84}$$

Using

$$\begin{aligned} \hat{z} \cdot \nabla \times (\mathbf{U}_1 \times \hat{z}) &= \nabla \cdot ((\mathbf{U}_1 \times \hat{z}) \times \hat{z}) \\ &= -\nabla \cdot \mathbf{U}_{1\perp} \end{aligned} \tag{4.85}$$

Eq. (4.84) becomes

$$\omega^2 \nabla_\perp \cdot \mathbf{U}_1 = k_\perp^2 \left(c_s^2 \nabla \cdot \mathbf{U}_1 + v_A^2 \nabla \cdot \mathbf{U}_{1\perp}\right) + k_z^2 v_A^2 \nabla \cdot \mathbf{U}_{1\perp}. \tag{4.86}$$

Equations (4.83) and (4.86) constitute two coupled equations in the variables $\nabla \cdot \mathbf{U}_1$ and $\nabla \cdot \mathbf{U}_{1\perp}$, namely

$$\begin{aligned} \left(\omega^2 - k^2 c_s^2\right) \nabla \cdot \mathbf{U}_1 - k^2 v_A^2 \nabla \cdot \mathbf{U}_{1\perp} &= 0 \\ k_\perp^2 c_s^2 \nabla \cdot \mathbf{U}_1 + \left(k^2 v_A^2 - \omega^2\right) \nabla \cdot \mathbf{U}_{1\perp} &= 0. \end{aligned} \tag{4.87}$$

These coupled equations have the determinant

$$\left(\omega^2 - k^2 c_s^2\right)\left(k^2 v_A^2 - \omega^2\right) + k^2 v_A^2 k_\perp^2 c_s^2 = 0, \tag{4.88}$$

which can be rearranged as a fourth order polynomial in ω,

$$\omega^4 - \omega^2 k^2 \left(v_A^2 + c_s^2\right) + k^2 k_z^2 v_A^2 c_s^2 = 0, \tag{4.89}$$

having roots

$$\omega^2 = \frac{k^2\left(v_A^2 + c_s^2\right) \pm \sqrt{k^4\left(v_A^2 + c_s^2\right)^2 - 4k^2 k_z^2 v_A^2 c_s^2}}{2}. \tag{4.90}$$

This dispersion relation has the following limiting forms

$$\left.\begin{aligned} &\omega^2 = k_\perp^2\left(v_A^2 + c_s^2\right) \\ &\text{or} \\ &\omega^2 = 0 \end{aligned}\right\} \text{ if } k_z = 0 \tag{4.91}$$

and

$$\left.\begin{aligned} &\omega^2 = k_z^2 v_A^2 \\ &\text{or} \\ &\omega^2 = k_z^2 c_s^2 \end{aligned}\right\} \text{ if } k_\perp^2 = 0; \tag{4.92}$$

the $\omega^2 = k_\perp^2\left(v_A^2 + c_s^2\right)$ mode existing in the $k_z = 0$ limit is called the magnetosonic mode and corresponds to choosing the plus sign in Eq. (4.90).

4.3.7 MHD shear (slow) mode

It is now assumed that $\nabla \cdot \mathbf{U}_1 = 0$ and taking the curl of Eq. (4.81) gives

$$\begin{aligned}
\frac{\partial^2 \nabla \times \mathbf{U}_1}{\partial t^2} &= v_A^2 \nabla \times \nabla \times \left(\frac{\partial \mathbf{U}_1}{\partial z} \times \hat{z} \right) \\
&= v_A^2 \nabla \times \left(\frac{\partial \mathbf{U}_1}{\partial z} \underbrace{\nabla \cdot \hat{z}}_{\text{zero}} + \hat{z} \cdot \nabla \frac{\partial \mathbf{U}_1}{\partial z} - \hat{z} \underbrace{\nabla \cdot \frac{\partial \mathbf{U}_1}{\partial z}}_{\text{zero}} - \frac{\partial \mathbf{U}_1}{\partial z} \cdot \underbrace{\nabla \hat{z}}_{\text{zero}} \right) \\
&= v_A^2 \frac{\partial^2}{\partial z^2} \nabla \times \mathbf{U}_1,
\end{aligned} \tag{4.93}$$

where the vector identity $\nabla \times (\mathbf{F} \times \mathbf{G}) = \mathbf{F} \nabla \cdot \mathbf{G} + \mathbf{G} \cdot \nabla \mathbf{F} - \mathbf{G} \nabla \cdot \mathbf{F} - \mathbf{F} \cdot \nabla \mathbf{G}$ has been used to obtain the second line.

Equation (4.93) reduces to the slow wave dispersion relation Eq. (4.68). The associated spatial behavior is such that $\nabla \times \mathbf{U}_1 \neq 0$, and the mode is unaffected by existence of finite pressure. The perturbed magnetic field is orthogonal to the equilibrium field, i.e., $\mathbf{B}_1 \cdot \mathbf{B} = 0$, since it has been assumed that $\nabla \cdot \mathbf{U}_1 = 0$ and since finite $\mathbf{B}_1 \cdot \mathbf{B}$ corresponds to finite $\nabla \cdot \mathbf{U}_1$. Since $\nabla \times \mathbf{U}_1$ is the fluid vorticity, the slow mode involves propagation of vorticity.

4.3.8 Limitations of the MHD model

The MHD model ignores parallel electron dynamics and so has a shear mode dispersion $\omega^2 = k_z^2 v_A^2$ that has no dependence on $k_\perp$. Some authors interpret this as a license to allow arbitrarily large $k_\perp$, in which case a shear mode could be localized to a single field line. However, the two-fluid model of the shear mode does have a dependence on $k_\perp$, which becomes important when $k_\perp$ becomes large. The nature of the two-fluid shear mode depends on how the parallel phase velocity of the wave compares to the electron and ion thermal velocities, i.e., on how $\omega/k_z \sim v_A$ compares to v_{Te} and v_{Ti}. Since it is possible to have (i) $v_A \ll v_{Ti}, v_{Te}$, (ii) $v_{Ti} \ll v_A \ll v_{Te}$, or (iii) $v_{Ti}, v_{Te} \ll v_A$, three distinct two-fluid regimes can exist and, as will be shown below, the plasma β determines which is the relevant regime for a given plasma.

MHD also predicts a sound wave that is identical to the ordinary hydrodynamic sound wave of an unmagnetized gas. The perpendicular behavior of this sound wave is consistent with the two-fluid model because both two-fluid and MHD perpendicular motions involve compressional behavior associated with having finite B_{z1}. However, the parallel behavior of the MHD sound wave is problematic because E_{z1} is assumed to be identically zero in MHD. According to the two-fluid model, any parallel acceleration requires a parallel electric field. The two-fluid B_{z1} mode is decoupled from the two-fluid E_{z1} mode so that the two-fluid B_{z1} mode is both compressional and has no parallel motion associated with it.

The MHD analysis makes no restriction on the electron to ion temperature ratio and predicts that a sound wave will exist when $T_e = T_i$. In contrast, the two-fluid model shows that sound waves can only exist when $T_e \gg T_i$ because only in this regime is it possible to have $\kappa T_i/m_i \ll \omega^2/k_z^2 \ll \kappa T_e/m_e$ and so have inertial behavior for ions and kinetic behavior for electrons.

Various paradoxes develop in the MHD treatment of the shear mode but not in the two-fluid description. These paradoxes illustrate the limitations of the MHD description of a plasma and show that MHD results must be treated with caution for the shear (slow) mode. MHD provides an adequate description for the fast (compressional) mode.

4.4 Two-fluid model of Alfvén modes

We now examine these modes from a two-fluid point of view. An important point that will be demonstrated is that the two-fluid point of view shows that the shear mode has three distinct forms depending on how v_A^2 compares to the ion and electron thermal velocities. The Alfvén velocity could be slower than both electron and ion thermal velocities, faster than the ion thermal velocity while slower than the electron thermal velocity, or faster than both the electron and ion thermal velocities. Which of these situations occurs depends upon the ratio of hydrodynamic pressure to magnetic pressure. This ratio is defined for each species σ as

$$\beta_\sigma = \frac{n\kappa T_\sigma}{B^2/\mu_0}; \tag{4.94}$$

the subscript σ is not used if electrons and ions have the same temperature. β_i measures the ratio of ion thermal velocity to the Alfvén velocity since

$$\frac{v_{Ti}^2}{v_A^2} = \frac{\kappa T_i/m_i}{B^2/nm_i\mu_0} = \beta_i. \tag{4.95}$$

Thus, $v_{Ti} \ll v_A$ corresponds to $\beta_i \ll 1$. Magnetic forces dominate hydrodynamic forces in a low-β plasma, whereas in a high-β plasma the opposite is true.

The ratio of electron thermal velocity to Alfvén velocity is

$$\frac{v_{Te}^2}{v_A^2} = \frac{\kappa T_e/m_e}{B^2/nm_i\mu_0} = \frac{m_i}{m_e}\beta_e. \tag{4.96}$$

Thus, $v_{Te}^2 \gg v_A^2$ when $\beta_e \gg m_e/m_i$ and $v_{Te}^2 \ll v_A^2$ when $\beta_e \ll m_e/m_i$. Shear Alfvén wave physics is different in the $\beta_e \gg m_e/m_i$ and $\beta_e \ll m_e/m_i$ regimes, which therefore must be investigated separately. MHD ignores this β_σ dependence and so oversimplifies these waves.

The MHD derivation used the polarization drift to give a relationship between $\mathbf{J}_{\perp 1}$ and $\mathbf{E}_{\perp 1}$ but was ambiguous regarding the relationship between J_{z1} and E_{z1}. The two-fluid model is based on the linearized equation of motion

$$m_\sigma n\frac{\partial \mathbf{u}_{\sigma 1}}{\partial t} = nq_\sigma\left(\mathbf{E}_1 + \mathbf{u}_{\sigma 1}\times\mathbf{B}\right) - \nabla\cdot\mathbf{P}_{\sigma 1}; \tag{4.97}$$

this equation gives essentially the same physics as MHD in the perpendicular direction, but not in the parallel direction. As in MHD, charge neutrality is assumed so that $n_i = n_e = n$. Also, to have increased generality, the pressure is allowed to be anisotropic so

$$\nabla\cdot\mathbf{P}_{\sigma 1} = \nabla\cdot\begin{bmatrix} P_{\sigma\perp 1} & 0 & 0 \\ 0 & P_{\sigma\perp 1} & 0 \\ 0 & 0 & P_{\sigma z1}\end{bmatrix} = \nabla_\perp P_{\sigma\perp 1} + \hat{z}\frac{\partial P_{\sigma z1}}{\partial z}. \tag{4.98}$$

Because quasi-neutrality is assumed, the perpendicular current given by two-fluid theory is essentially identical to MHD and consists of ion polarization drift and diamagnetic drift. The existence of these drifts can be seen by invoking the $\omega \ll \omega_{c\sigma}$ assumption so that the left-hand side of Eq. (4.97) can be neglected to first approximation, in which case the perpendicular component of Eq. (4.97) can be expressed as

$$\mathbf{u}_{\sigma 1}\times\mathbf{B} \simeq -\mathbf{E}_{\perp 1} + \nabla_\perp P_{\sigma\perp 1}/nq_\sigma. \tag{4.99}$$

Crossing with $\mathbf{B}$ to solve for $\mathbf{u}_{\sigma\perp 1}$ gives

$$\mathbf{u}_{\sigma\perp 1} = \frac{\mathbf{E}_1\times\mathbf{B}}{B^2} - \frac{\nabla P_{\sigma\perp 1}\times\mathbf{B}}{nq_\sigma B^2}, \tag{4.100}$$

where the first term is the single-particle $\mathbf{E}\times\mathbf{B}$ drift and the second term is the diamagnetic drift, a fluid effect. Because $\mathbf{u}_{\sigma\perp 1}$ is time-dependent the polarization drift, given in Eq. (4.50), should also appear. Polarization drift results from solving the perpendicular equation of motion to first order in $\omega/\omega_{c\sigma}$. Recalling that the form of the single-particle polarization drift is $\mathbf{v}_p = \left(m_\sigma/q_\sigma B^2\right)\mathrm{d}\mathbf{E}_{\perp 1}/\mathrm{d}t$ and then using $\mathbf{E}_{\perp 1} - \nabla_\perp P_{\sigma\perp 1}/nq_\sigma$ for the fluid model being considered here instead of just $\mathbf{E}_{\perp 1}$ for single particles (cf. right-hand side of Eq. (4.99)), the fluid polarization drift is obtained. With the inclusion of this higher order correction, the perpendicular fluid motion becomes

$$\mathbf{u}_{\sigma\perp 1} = \frac{\mathbf{E}_1\times\mathbf{B}}{B^2} - \frac{\nabla P_{\sigma\perp 1}\times\mathbf{B}}{nq_\sigma B^2} + \frac{m_\sigma}{q_\sigma B^2}\frac{\mathrm{d}\mathbf{E}_{\perp 1}}{\mathrm{d}t} - \frac{m_\sigma}{nq_\sigma^2 B^2}\nabla_\perp\frac{\mathrm{d}P_{\sigma\perp 1}}{\mathrm{d}t}. \tag{4.101}$$

The $\mathrm{d}\mathbf{E}_{\perp 1}/\mathrm{d}t$ and $\mathrm{d}P_{\sigma\perp 1}/\mathrm{d}t$ terms are respectively smaller than the corresponding $\mathbf{E}_1$ and $P_{\sigma\perp 1}$ terms by the ratio $\omega/\omega_{c\sigma}$ and so may be ignored when electron and

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ion fluid velocities are considered separately because $\omega \ll \omega_{c\sigma}$ by assumption. However, when calculating the perpendicular current, i.e., $\mathbf{J}_{\perp 1} = \sum nq_\sigma \mathbf{u}_{\sigma\perp 1}$, the electron and ion $\mathbf{E}\times\mathbf{B}$ drift terms cancel each other so that the polarization terms become the leading terms involving the electric field. Because of the mass in the numerator, the ion polarization drift is much larger than the electron polarization drift. Thus, the perpendicular current comes from ion polarization drift and diamagnetic current

$$\mu_0 \mathbf{J}_{\perp 1} = \frac{\mu_0 n m_i \dot{\mathbf{E}}_{\perp 1}}{B^2} - \sum_\sigma \frac{\nabla P_{\sigma\perp 1}\times\mathbf{B}}{B^2} = \frac{1}{v_A^2}\dot{\mathbf{E}}_{\perp 1} - \frac{\mu_0 \nabla P_{\perp 1}\times\mathbf{B}}{B^2}, \tag{4.102}$$

where $P_{\perp 1} = \sum P_{\sigma\perp 1}$ and the dot on top of $\mathbf{E}_{\perp 1}$ denotes time derivative. The term involving $\dot{P}_{\perp 1}$ has been dropped because it is small by ω/ω_c compared to the $P_{\perp 1}$ term. Equation (4.102) is essentially the same as what one would obtain by crossing the MHD equation of motion with $\mathbf{B}$ and assuming that the MHD velocity $\mathbf{U}$ is given by the $\mathbf{E}\times\mathbf{B}$ drift. Thus, the two-fluid perpendicular dynamics is essentially the same as MHD perpendicular dynamics.

We now reconsider equations of the form given by Eqs. (4.57) and (4.58), but do so without assuming that E_{z1} is zero. Thus, all vectors are decomposed into components parallel and perpendicular to the equilibrium magnetic field, e.g., $\mathbf{E}_1 = \mathbf{E}_{\perp 1} + E_{z1}\hat{z}$. The ∇ operator is similarly decomposed into components parallel to and perpendicular to the equilibrium magnetic field, i.e., $\nabla = \nabla_\perp + \hat{z}\partial/\partial z$, and all quantities are assumed to be proportional to $f(x,y)\exp(\mathrm{i}k_z z - \mathrm{i}\omega t)$. Faraday's law can then be written as

$$\nabla_\perp \times \mathbf{E}_{\perp 1} + \nabla_\perp \times E_{z1}\hat{z} + \hat{z}\frac{\partial}{\partial z}\times\mathbf{E}_{\perp 1} = -\frac{\partial}{\partial t}\left(\mathbf{B}_{\perp 1} + B_{z1}\hat{z}\right), \tag{4.103}$$

which has a parallel component

$$\hat{z}\cdot\nabla_\perp \times \mathbf{E}_{\perp 1} = \mathrm{i}\omega B_{z1} \tag{4.104}$$

and a perpendicular component

$$\left(\nabla_\perp E_{z1} - \mathrm{i}k_z \mathbf{E}_{\perp 1}\right)\times\hat{z} = \mathrm{i}\omega\mathbf{B}_{\perp 1}. \tag{4.105}$$

Similarly, Ampère's law can be decomposed into a parallel component

$$\hat{z}\cdot\nabla_\perp \times \mathbf{B}_{\perp 1} = \mu_0 J_{z1} \tag{4.106}$$

and a perpendicular component

$$\left(\nabla_\perp B_{z1} - \mathrm{i}k_z \mathbf{B}_{\perp 1}\right)\times\hat{z} = \mu_0\mathbf{J}_{\perp 1}. \tag{4.107}$$

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Unlike the MHD model, here we have taken into account both finite E_{z1} and the parallel component of Ampère's law. Substituting the perpendicular current given by Eq. (4.102) into Eq. (4.107) gives

$$\left(\nabla_\perp B_{z1} - \mathrm{i}k_z \mathbf{B}_{\perp 1}\right)\times\hat{z} = -\frac{\mathrm{i}\omega}{v_A^2}\mathbf{E}_{\perp 1} - \frac{\mu_0 \nabla P_1 \times \hat{z}}{B} \tag{4.108}$$

or, after rearrangement,

$$\nabla_\perp\left(B_{z1} + \frac{\mu_0 P_{\perp 1}}{B}\right)\times\hat{z} - \mathrm{i}k_z\mathbf{B}_{\perp 1}\times\hat{z} = -\frac{\mathrm{i}\omega}{v_A^2}\mathbf{E}_{\perp 1}. \tag{4.109}$$

Equations (4.105) and (4.109) can be considered as two coupled equations involving $\mathbf{E}_{\perp 1}$ and $\mathbf{B}_{\perp 1}$. After crossing Eq. (4.105) with $\hat{z}$, they can also be written as two coupled equations involving $\mathbf{E}_{\perp 1}$ and $\mathbf{B}_{\perp 1}\times\hat{z}$, namely

$$\mathrm{i}\omega\mathbf{B}_{\perp 1}\times\hat{z} - \mathrm{i}k_z\mathbf{E}_{\perp 1} = -\nabla_\perp E_{z1} \tag{4.110a}$$

$$-\mathrm{i}k_z\mathbf{B}_{\perp 1}\times\hat{z} + \frac{\mathrm{i}\omega}{v_A^2}\mathbf{E}_{\perp 1} = -\nabla_\perp\left(B_{z1} + \frac{\mu_0 P_{\perp 1}}{B}\right)\times\hat{z}. \tag{4.110b}$$

These may now be solved algebraically for $\mathbf{E}_{\perp 1}$ and $\mathbf{B}_{\perp 1}\times\hat{z}$ to obtain

$$\mathbf{E}_{\perp 1} = \frac{\mathrm{i}k_z\nabla_\perp E_{z1} + \mathrm{i}\omega\nabla_\perp\left(B_{z1} + \mu_0 P_{\perp 1}/B\right)\times\hat{z}}{\omega^2/v_A^2 - k_z^2} \tag{4.111a}$$

$$\mathbf{B}_{\perp 1}\times\hat{z} = \frac{\mathrm{i}\left(\omega/v_A^2\right)\nabla_\perp E_{z1} + \mathrm{i}k_z\nabla_\perp\left(B_{z1} + \mu_0 P_{\perp 1}/B\right)\times\hat{z}}{\omega^2/v_A^2 - k_z^2}. \tag{4.111b}$$

Equations (4.104) and (4.106), the respective parallel components of Faraday's and Ampère's laws, can be written as

$$\nabla\cdot(\mathbf{E}_{\perp 1}\times\hat{z}) = \mathrm{i}\omega B_{z1}, \tag{4.112a}$$

$$\nabla\cdot(\mathbf{B}_{\perp 1}\times\hat{z}) = \mu_0 J_{z1}, \tag{4.112b}$$

so substituting for $\mathbf{E}_{\perp 1}\times\hat{z}$ and $\mathbf{B}_{\perp 1}\times\hat{z}$ gives

$$\nabla\cdot\left(\frac{\mathrm{i}k_z\nabla_\perp E_{z1}\times\hat{z} - \mathrm{i}\omega\nabla_\perp\left(B_{z1} + \mu_0 P_{\perp 1}/B\right)}{\omega^2/v_A^2 - k_z^2}\right) = \mathrm{i}\omega B_{z1} \tag{4.113a}$$

$$\nabla\cdot\left(\frac{\mathrm{i}\left(\omega/v_A^2\right)\nabla_\perp E_{z1} + \mathrm{i}k_z\nabla_\perp\left(B_{z1} + \mu_0 P_{\perp 1}/B\right)\times\hat{z}}{\omega^2/v_A^2 - k_z^2}\right) = \mu_0 J_{z1}. \tag{4.113b}$$

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Because $\nabla\cdot\left(\nabla_\perp E_{z1}\times\hat{z}\right)=0$ and $\nabla\cdot\left(\nabla_\perp\left(B_{z1}+\mu_0 P_{\perp 1}/B\right)\times\hat{z}\right)=0$ by virtue of the vector identity $\nabla\cdot(\nabla_\perp\psi\times\hat{z})=\nabla\cdot(\nabla\psi\times\hat{z})=\hat{z}\cdot\nabla\times\nabla\psi=0$, these equations simplify to

$$\nabla\cdot\left(\frac{\nabla_\perp\left(B_{z1}+\mu_0 P_{\perp 1}/B\right)}{\omega^2/v_A^2-k_z^2}\right)=-B_{z1} \tag{4.114a}$$

$$\nabla\cdot\left(\frac{\mathrm{i}\omega\nabla_\perp E_{z1}}{\omega^2/v_A^2-k_z^2}\right)=v_A^2\mu_0 J_{z1}. \tag{4.114b}$$

Equation (4.114a) is essentially the equation for the compressional mode, but some further effort is required to relate $P_{\perp 1}$ to B_{z1} in order to establish this. Equation (4.114b) is essentially the equation for the shear mode, but here some further effort is required to establish the relationship between J_{z1} and E_{z1}.

An important property distinguishing these modes is whether or not the motion in the perpendicular direction involves compression. To demonstrate this, we take the divergence of Eq. (4.100) (the small polarization drift terms included in the extended form given in Eq. (4.101) are ignored since these polarization terms are higher order in $\omega/\omega_{c\sigma}$ and are important only when calculating perpendicular current). The result is

$$\nabla\cdot\mathbf{u}_{\sigma\perp 1}=\frac{1}{B^2}\mathbf{B}\cdot\nabla\times\mathbf{E}_1=-\frac{1}{B^2}\mathbf{B}\cdot\frac{\partial\mathbf{B}_1}{\partial t}=\frac{\mathrm{i}\omega}{B}B_{z1}. \tag{4.115}$$

Thus, modes where $B_{z1}=0$ involve no compression in the perpendicular direction; these are the shear modes. The modes where B_{z1} is finite are the compressional modes (finite B_{z1} leads to finite $\nabla\cdot\mathbf{u}_{\sigma\perp 1}$, which leads to finite $P_{\perp 1}$ as the plasma is squeezed in the perpendicular direction).

4.4.1 Two-fluid slow (shear) modes

The two-fluid shear modes fall into three categories depending on how the Alfvén velocity compares to the electron and ion thermal velocities. These three regimes are shown schematically in Fig. 4.3. The shear modes have $B_{z1}=0$ so $\nabla\cdot\mathbf{u}_{\sigma\perp 1}=0$. The parallel component of the linearized equation of motion, Eq. (4.97), is

$$nm_\sigma\frac{\partial u_{\sigma z1}}{\partial t}=nq_\sigma E_{z1}-\gamma_\sigma\kappa T_{\|\sigma}\frac{\partial n_{\sigma 1}}{\partial z}, \tag{4.116}$$

where $P_{\sigma\|1}=\gamma_\sigma n_{\sigma 1}\kappa T_{\|\sigma}$ has been used. We choose $\gamma=1$ if the motion is isothermal and $\gamma_\sigma=3$ if the motion is adiabatic since the compression, being parallel, is one-dimensional. The isothermal case corresponds to $\omega^2/k_z^2\ll\kappa T_\sigma/m_\sigma$ and vice versa for the adiabatic case.

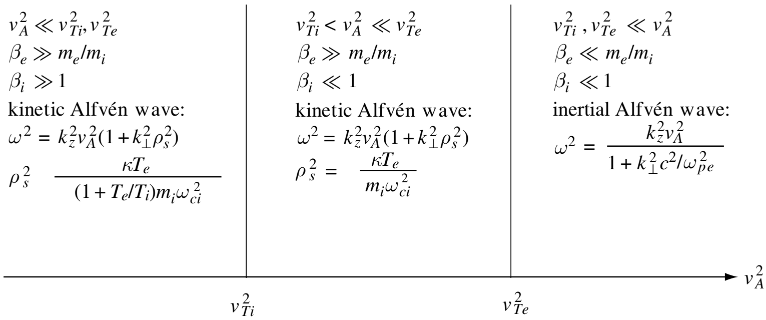

Fig. 4.3 The two-fluid shear mode dispersion relation depends on how v_A^2 compares to the electron and ion thermal velocities and this comparison depends on the electron and ion β's.

Because $\nabla \cdot \mathbf{u}_{\sigma\perp 1} = 0$ the continuity equation is

$$\frac{\partial n_{\sigma 1}}{\partial t} + \frac{\partial}{\partial z}\left(n u_{\sigma z1}\right) = 0. \tag{4.117}$$

Taking the time derivative of Eq. (4.116) and invoking Eq. (4.117) gives

$$\frac{\partial^2 u_{\sigma z1}}{\partial t^2} - \gamma_\sigma \frac{\kappa T_\sigma}{m_\sigma}\frac{\partial^2 u_{\sigma z1}}{\partial z^2} = \frac{q_\sigma}{m_\sigma}\frac{\partial E_{z1}}{\partial t}, \tag{4.118}$$

which is similar to electron plasma wave and ion acoustic wave dynamics except it has *not* been assumed that E_{z1} is electrostatic.

Invoking the assumption that all quantities are of the form $f(x, y)\exp(\mathrm{i}k_z z - \mathrm{i}\omega t)$ Eq. (4.118) can be solved to give

$$u_{\sigma z1} = \frac{\mathrm{i}\omega q_\sigma}{m_\sigma}\frac{E_{z1}}{\omega^2 - \gamma_\sigma k_z^2 \kappa T_\sigma/m_\sigma} \tag{4.119}$$

and so the sought-after relation between parallel current and parallel electric field is

$$\mu_0 J_{z1} = \frac{\mathrm{i}\omega}{c^2} E_{z1} \sum_\sigma \frac{\omega_{p\sigma}^2}{\omega^2 - \gamma_\sigma k_z^2 \kappa T_\sigma/m_\sigma}. \tag{4.120}$$

This provides the prescription for J_{z1} needed in the right-hand side of Eq. (4.114b) and contains the parallel dynamics physics omitted from the MHD description. Substitution of Eq. (4.120) into the right-hand side of Eq. (4.114b) gives the differential equation for the shear wave

$$\nabla \cdot \left(\frac{\nabla_\perp E_{z1}}{\omega^2/v_A^2 - k_z^2}\right) - \frac{v_A^2}{c^2} E_{z1} \sum_\sigma \frac{\omega_{p\sigma}^2}{\omega^2 - \gamma_\sigma k_z^2 \kappa T_\sigma/m_\sigma} = 0. \tag{4.121}$$

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On replacing $\nabla_\perp \to \mathrm{i}\mathbf{k}_\perp$, Eq. (4.121) becomes

$$\frac{k_\perp^2}{\omega^2 - k_z^2 v_A^2} + \frac{\omega_{pe}^2}{c^2}\frac{1}{\omega^2 - \gamma_e k_z^2 \kappa T_e/m_e} + \frac{\omega_{pi}^2}{c^2}\frac{1}{\omega^2 - \gamma_i k_z^2 \kappa T_i/m_i} = 0. \tag{4.122}$$

We are interested in the solution to Eq. (4.122) where $\omega^2/k_z^2 \sim v_A^2$ but need to take into account three different possibilities for how v_A^2 might compare to electron and ion thermal velocities, namely (i) $v_A^2 \gg v_{Te}^2, v_{Ti}^2$, (ii) $v_{Te}^2 \gg v_A^2 \gg v_{Ti}^2$, or (iii) $v_{Te}^2, v_{Ti}^2 \gg v_A^2$. These three possibilities correspond respectively to the following β situations: (i) $\beta_e \ll m_e/m_i$, (ii) $m_i\beta_e/m_e \gg 1 \gg \beta_i$, and (iii) $\beta_i \gg 1$. These three situations are shown schematically in Fig. 4.3 and will now be discussed in detail.

In situation (i) where $\omega^2/k_z^2 \gg \kappa T_e/m_e$, the second term dominates the third term since $\omega_{pe}^2 \gg \omega_{pi}^2$ and so Eq. (4.122) can be recast as

$$\omega^2 = \frac{k_z^2 v_A^2}{1 + k_\perp^2 c^2/\omega_{pe}^2}, \tag{4.123}$$

which is called the inertial Alfvén wave (IAW). The quantity c/ω_{pe} is called the electron collisionless skin depth. If $k_\perp^2 c^2/\omega_{pe}^2$ is not too large, then ω/k_z is of the order of the Alfvén velocity and the condition $\omega^2 \gg k_z^2 \kappa T_e/m_e$ corresponds to $v_A^2 \gg \kappa T_e/m_e$ or

$$\beta_e = \frac{n\kappa T_e}{B^2/\mu_0} \ll \frac{m_e}{m_i}. \tag{4.124}$$

Thus, inertial Alfvén wave shear modes exist only in the ultra-low β regime where $\beta_e \ll m_e/m_i$. The inertial Alfvén wave is called a cold plasma wave because its dispersion relation does not depend on temperature.

In situation (ii) where $\kappa T_i/m_i \ll \omega^2/k_z^2 \ll \kappa T_e/m_e$, Eq. (4.122) can be recast as

$$\frac{k_\perp^2}{\omega^2 - k_z^2 v_A^2} - \frac{\omega_{pe}^2}{c^2}\frac{1}{k_z^2 \kappa T_e/m_e} + \frac{\omega_{pi}^2}{c^2}\frac{1}{\omega^2} = 0. \tag{4.125}$$

Because ω^2 appears in the respective denominators of two distinct terms, Eq. (4.125) is fourth order in ω^2 and so describes two distinct modes. Let us suppose that the mode in question is much faster than the ion acoustic velocity, i.e., $\omega^2/k_z^2 \gg \kappa T_e/m_i$. In this case the last term can be dropped and the remaining two terms can be rearranged to give

$$\omega^2 = k_z^2 v_A^2 \left(1 + \frac{k_\perp^2}{v_A^2}\frac{\kappa T_e}{m_e}\frac{c^2}{\omega_{pe}^2}\right); \tag{4.126}$$

this is called the kinetic Alfvén wave (KAW). By defining

$$\rho_s^2 = \frac{1}{v_A^2}\frac{\kappa T_e}{m_e}\frac{c^2}{\omega_{pe}^2} = \frac{\kappa T_e}{m_i \omega_{ci}^2} \tag{4.127}$$

as a fictitious ion Larmor radius calculated using the electron temperature instead of the ion temperature, the kinetic Alfvén wave (KAW) dispersion relation can be expressed more succinctly as

$$\omega^2 = k_z^2 v_A^2 \left(1 + k_\perp^2 \rho_s^2\right). \tag{4.128}$$

If $k_\perp^2 \rho_s^2$ is not too large, then ω/k_z is again of the order of v_A and so the condition $\omega^2 \ll k_z^2 \kappa T_e/m_e$ corresponds to having $\beta_e \gg m_e/m_i$. The condition $\omega^2/k_z^2 \gg \kappa T_e/m_i$, which was also assumed, corresponds to assuming that $\beta_e \ll 1$. Thus, the KAW dispersion relation Eq. (4.128) is valid in the regime $m_e/m_i \ll \beta_e \ll 1$.

Let us now consider situation (iii) where $\omega^2/k_z^2 \ll \kappa T_i/m_i,\ \kappa T_e/m_e$. In this case Eq. (4.122) again reduces to

$$\omega^2 = k_z^2 v_A^2 \left(1 + k_\perp^2 \rho_s^2\right) \tag{4.129}$$

but now ρ_s^2 is defined as

$$\rho_s^2 = \frac{\omega_{pi}^2}{\omega_{ci}^2}\left(\frac{1}{\lambda_{De}^2} + \frac{1}{\lambda_{Di}^2}\right)^{-1} = \frac{\kappa T_e}{\left(1 + T_e/T_i\right) m_i \omega_{ci}^2}. \tag{4.130}$$

This situation would describe shear modes in a high-β plasma (ion thermal velocity faster than Alfvén velocity).

To summarize: the shear mode has $B_{z1} = 0,\ E_{z1} \neq 0,\ J_{z1} \neq 0$, and exists as a cold plasma inertial Alfvén wave for $\beta_e \ll m_e/m_i$ and as one of two types of warm plasma kinetic Alfvén waves for $\beta_e \gg m_e/m_i$. These are shown in Fig. 4.3. The shear mode involves incompressible perpendicular motion, i.e., $\nabla \cdot \mathbf{u}_{\sigma\perp 1} = i\mathbf{k}_\perp \cdot \mathbf{u}_{\sigma\perp 1} = 0$, which means that $\mathbf{k}_\perp$ is orthogonal to $\mathbf{u}_{\sigma\perp 1}$. For example, in Cartesian geometry, this means that if $\mathbf{u}_{\sigma\perp 1}$ is in the x direction, then $\mathbf{k}_\perp$ must be in the y direction, while in cylindrical geometry, this means that if $\mathbf{u}_{\sigma\perp 1}$ is in the θ direction, then $\mathbf{k}_\perp$ must be in the r direction. The inertial Alfvén wave is known as a cold plasma wave because its dispersion relation does not depend on temperature (such a mode would exist even in the limit of a cold plasma). The kinetic Alfvén wave depends on the plasma having finite temperature and is therefore called a warm plasma wave. The shear mode parallel dynamics is related to parallel propagating ion acoustic modes since both shear Alfvén waves and parallel propagating ion acoustic modes involve parallel motion driven by finite E_{z1}; this parallel physics is missing from the MHD description.

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4.4.2 Two-fluid compressional modes

The compressional mode has $B_{z1} \neq 0$ and $E_{z1} = 0$. Having $E_{z1} = 0$ means $J_{z1} = 0$ and, since there is no parallel motion, the linearized continuity equation becomes

$$\frac{\partial n_{\sigma 1}}{\partial t} + n\nabla \cdot \mathbf{u}_{\sigma \perp 1} = 0. \tag{4.131}$$

Using Eq. (4.115) this may be expressed as

$$\frac{\partial n_{\sigma 1}}{\partial t} - \frac{n}{B}\frac{\partial B_{z1}}{\partial t} = 0, \tag{4.132}$$

which may be integrated with respect to time to give

$$\frac{n_{\sigma 1}}{n} = \frac{B_{z1}}{B}. \tag{4.133}$$

Assuming an adiabatic response for this perpendicular compression gives

$$\frac{P_{\perp 1}}{P} = \gamma \frac{n_1}{n} = \gamma \frac{B_{z1}}{B}. \tag{4.134}$$

Substitution for $P_{\perp 1}$ in Eq. (4.114a) gives

$$\nabla_\perp \cdot \left(\frac{\left(v_A^2 + c_s^2\right)}{\omega^2 - k_z^2 v_A^2} \nabla_\perp B_{z1} \right) + B_{z1} = 0, \tag{4.135}$$

where

$$c_s^2 = \gamma \kappa \frac{T_e + T_i}{m_i}. \tag{4.136}$$

On replacing $\nabla_\perp \to \mathrm{i}\mathbf{k}_\perp$, Eq. (4.135) becomes the compressional mode dispersion relation

$$\frac{-k_\perp^2 \left(v_A^2 + c_s^2\right)}{\omega^2 - k_z^2 v_A^2} + 1 = 0 \tag{4.137}$$

or

$$\omega^2 = k^2 v_A^2 + k_\perp^2 c_s^2, \tag{4.138}$$

where $k^2 = k_z^2 + k_\perp^2$.

4.5 Assignments

1. Plot frequency versus wavenumber for both the electron plasma wave and the ion acoustic wave in an unmagnetized argon plasma, which has $n = 10^{18}$ m^{-3}, $T_e = 10$ eV, and $T_i = 1$ eV.

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2. Let $\Delta\phi$ be the difference between the phase shift a 632.8 nm helium–neon laser beam experiences on traversing a given length of vacuum and on traversing the same length of plasma. What is $\Delta\phi$ when the laser beam passes through 10 cm of plasma having a density of $n = 10^{22}$ m^3? How could this be used as a density diagnostic?
3. Prove that the electrostatic plasma wave $\omega^2 = \omega_{pe}^2 + 3k^2\kappa T_e/m_e$ can also be written as

$$\omega^2 = \omega_{pe}^2(1 + 3k^2\lambda_{De}^2)$$

and show over what range of $k^2\lambda_{De}^2$ the dispersion is valid. Plot the dispersion $\omega(k)$ versus k for both negative and positive k. Next, plot on the same graph the electromagnetic dispersion $\omega^2 = \omega_{pe}^2 + k^2c^2$ and show the limits of validity. Plot the ion acoustic dispersion $\omega^2 = k^2c_s^2/(1 + k^2\lambda_{De}^2)$ on this graph showing its region of validity. Finally, plot the ion acoustic dispersion with a finite ion temperature. Show the limits of validity of the ion acoustic dispersion.
4. Physical picture of plasma oscillations. Suppose that a plasma is cold and initially neutral. Consider a spherical volume of this plasma and imagine that a thin shell of electrons at spherical radius r having thickness δr moves radially outward by a distance equal to its thickness. Suppose further that the ions are infinitely massive and cannot move. What is the total ion charge acting on the electrons (consider the charge density and volume of the ions left behind when the electron shell is moved out)? What is the electric field due to these ions? By considering the force due to this electric field on an individual electron in the shell, show that the entire electron shell will execute simple harmonic motion at the frequency ω_{pe}. If the ions had finite mass how would you expect the problem to be modified (hint: consider the reduced mass)?
5. Suppose that an MHD plasma immersed in a uniform magnetic field $\mathbf{B} = B_0\hat{z}$ has an oscillating electric field $\tilde{\mathbf{E}}_\perp$, where $\perp$ means in the direction perpendicular to $\hat{z}$. What is the polarization current associated with $\tilde{\mathbf{E}}_\perp$? By substituting this polarization current into the MHD approximation of Ampère's law, find a relationship between $\partial\tilde{\mathbf{E}}_\perp/\partial t$ and a spatial operator on $\tilde{\mathbf{B}}$. Use Faraday's law to obtain a similar relationship between $\partial\tilde{\mathbf{B}}_\perp/\partial t$ and a spatial operator on $\tilde{\mathbf{E}}$. Consider a mode where $\tilde{E}_x(z,t)$ and $\tilde{B}_y(z,t)$ are the only finite components and derive a wave equation. Do the same for the pair $\tilde{E}_y(x,t)$ and $\tilde{B}_z(x,t)$. Which mode is the compressional mode and which is the shear mode?